

The blackboard game, explained from scratch

What the agents do, what the controller does, which pools we chose, and how the language-model-free simulation mirrors it

MA-CC blackboard-control study

4 October 2026

1 The task, in plain terms

Three people must be split between two jobs: one works alone on job 1, the other two work together on job 2. There are therefore exactly three ways to do it, called `ALLOCATION_0`, `ALLOCATION_1` and `ALLOCATION_2`, one for each choice of the lone worker.

Nine hidden numbers, each 1, 2 or 3, describe the world: for each person a skill at each job (six numbers), plus how well each pair cooperates (three numbers). An allocation’s score is

$$\text{score} = \underbrace{\text{lone worker's skill at job 1}}_{\text{one number}} + \underbrace{\text{the pair's two skills at job 2}}_{\text{two numbers}} + \underbrace{\text{how well the pair cooperates}}_{\text{one number}}.$$

The highest score wins. In our fixed world the hidden numbers are (3, 1, 1, 2, 2, 1, 1, 1, 2), the scores are (8, 4, 6), so `ALLOCATION_0` is **correct** and `ALLOCATION_2` — the runner-up — is the wrong answer a hostile controller tries to promote.

Facts. A *fact* is a true sentence about the world, drawn from a fixed list of 99 possible sentences (“person p is at least moderately skilled at job j ”, “ p is better than p' at job j ”). Exactly **49 of them are true** in our world, and only those can ever be said. **Nobody ever lies.** All the steering is done by choosing *which* true things to say.

Belief. Nobody sees the nine hidden numbers. But because there are only 3^9 possibilities, we can enumerate them: 14,388 of them have a unique winner. An agent holding a set K of facts believes

$$\mathbb{P}(a \mid K) = \frac{\#\{\text{worlds consistent with } K \text{ where } a \text{ wins}\}}{\#\{\text{worlds consistent with } K\}}, \quad (1)$$

which is a *count*, not an estimate. With no facts at all this is 1/3 for each allocation — the *prior*.

2 How the full language-model simulation runs

What the agents do. Each of the 24 agents starts holding **exactly one** fact — not a packet. Every round:

1. **Forget.** Each fact an agent currently uses survives with probability ρ (*persistence*). At $\rho = 1$ nothing is forgotten. Forgotten facts are not erased, they only stop being used.
2. **Read.** The agent reads messages on the board. **Any fact it reads becomes usable by it.** This is the only way facts spread.
3. **Think and vote.** It computes Eq. (1) from the facts it can currently use, and votes.
4. **Post.** It writes one message citing one fact it holds.

Agents are updated **one at a time**, so the twelfth agent sees the posts of the first eleven. The board is wiped at the end of each round.

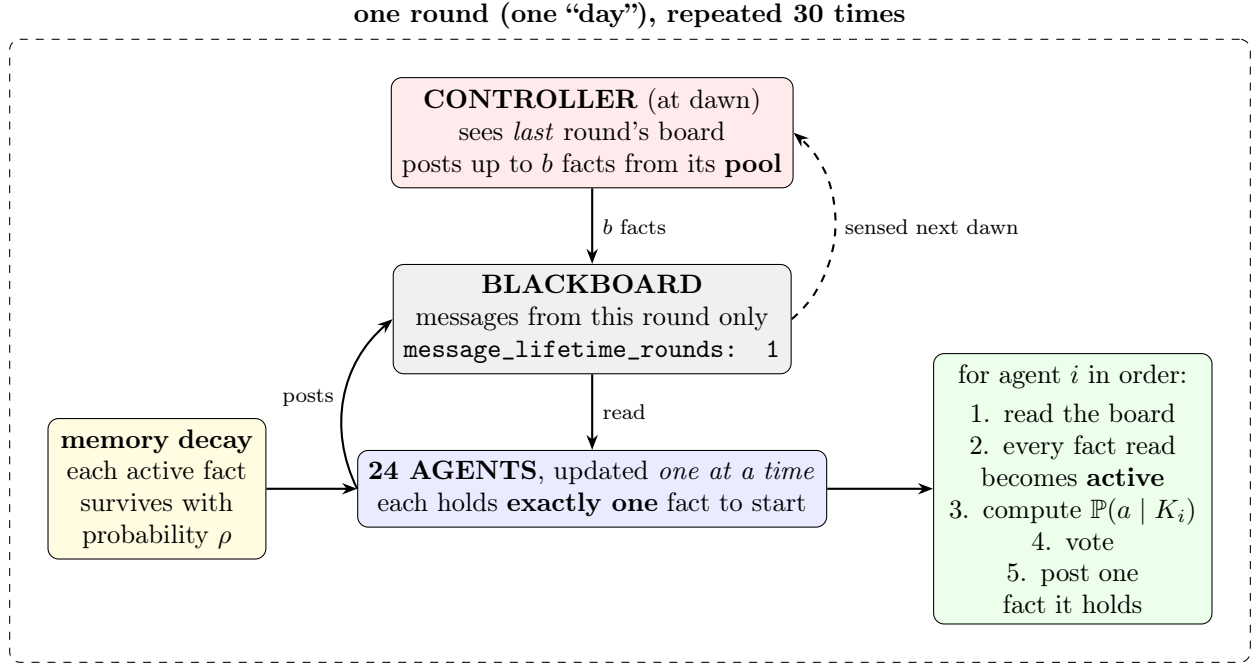


Figure 1: One round of the game. The dashed arrow is the *feedback*: the controller’s next action depends on what the population did.

What the controller does. It is an extra participant with its own small set of facts — its **pool**. Once per round, at dawn: it looks at last round’s board, decides whether to act, and if so posts up to b facts from its pool. It is given a *target*: **ALLOCATION_0** (the truth) or **ALLOCATION_2** (the wrong answer). It never lies either; it only chooses what to mention.

The modes that exist.

knob	our choice	what the alternatives mean
agents’ communication	report_only	full_communication also lets agents post <i>re-requests</i> , which carry no fact
board reading	uniform , $q=7$	full reads every message — but see §4, it removes all randomness at $\rho=1$
controller action type	adaptive_communication	only this and truthful_strategic_report can post a pool fact at all
who writes it	llm_authored	deterministic uses a fixed rule — and §5 shows that rule cannot steer
side channel	recommendation_only	recommendation_plus_fact adds one fixed fact per episode through a private channel
budget	$b = 3$ per round	an <i>episode</i> reservoir B lets the controller choose <i>when</i> to spend
gate	always	soft only fires when the population is on the wrong side of a threshold

3 The two setups we designed, and their pools

Both use the *same* world and the *same* 49 facts. They differ only in **which facts the 24 agents hold**.

	symmetric-v2	nosolution-v2
agents	24	24
distinct facts held	16	15
how many agents lean each way (A0/A1/A2)	8 / 8 / 8	8 / 8 / 8
belief if you pooled every agent's facts	(1.000, 0, 0)	(0.500, 0, 0.500)
can the population prove the right answer?	yes	no
controller pool (shared by both targets)	12 facts	12 facts
pool composition (A0/A1/A2)	4 / 4 / 4	4 / 4 / 4
belief from the whole pool	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	(0.318, 0.364, 0.318)
does the pool prove anything?	no	no
overlap between pool and agents' facts	0	0

Why one shared pool. If the truth controller and the false controller drew from *different* pools, any difference in outcome could be the pool rather than the policy. One pool, identical for both, means every directional difference is attributable to *what the controller chooses to say*.

The plan. Three arms — no control, truth control, false control — crossed with $\rho \in \{0.75, 1\}$ and the two setups: **12 cells**, 100 episodes of 30 rounds each. Later, one factor at a time: the language-model controller, full communication, the episode budget, partial sensing.

4 The language-model-free simulation

The same mechanism, with one substitution: wherever the real game asks a language model to decide, we compute Eq. (1) exactly and take the most likely allocation. No words, no tuned numbers, **no free parameters**. A grid that would take a language model roughly 288 hours runs in **28 seconds**.

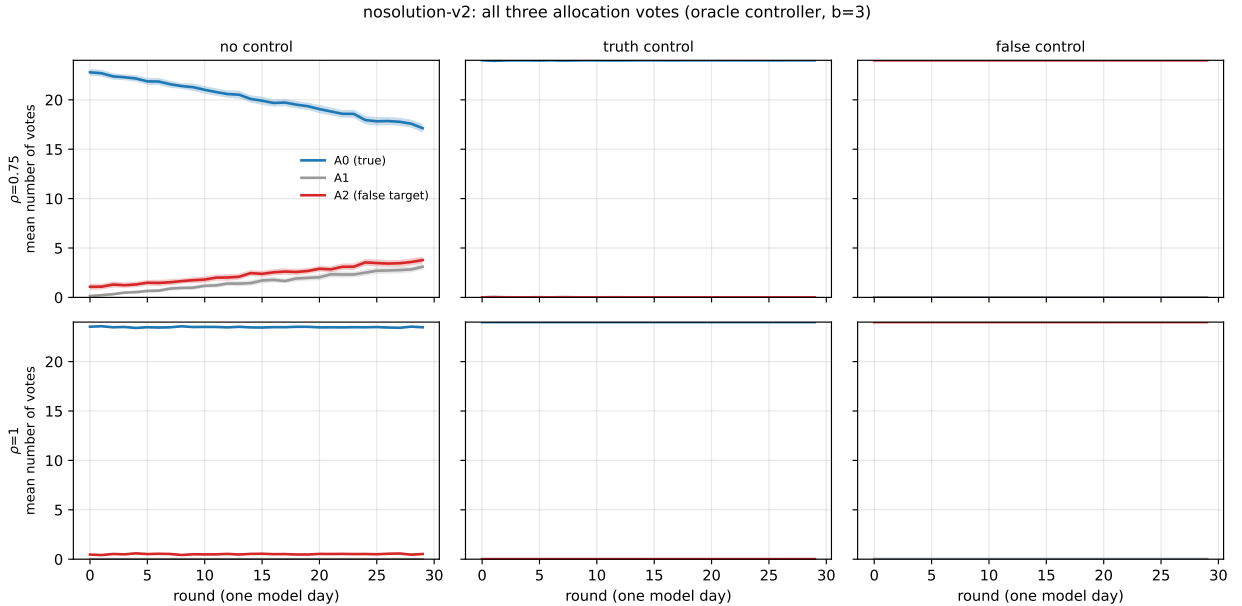


Figure 2: nosolution-v2: mean vote counts per round, in the style of the task_004 report. Bands are 95% intervals. The false controller takes the population from a near-tie to a large **ALLOCATION_2** majority.

5 Two things the simulation taught us

The repository's deterministic controller cannot steer. Its rule ranks the pool by *how long ago each fact was last posted*; the target enters only as a tie-break, and with a neutral pool it never

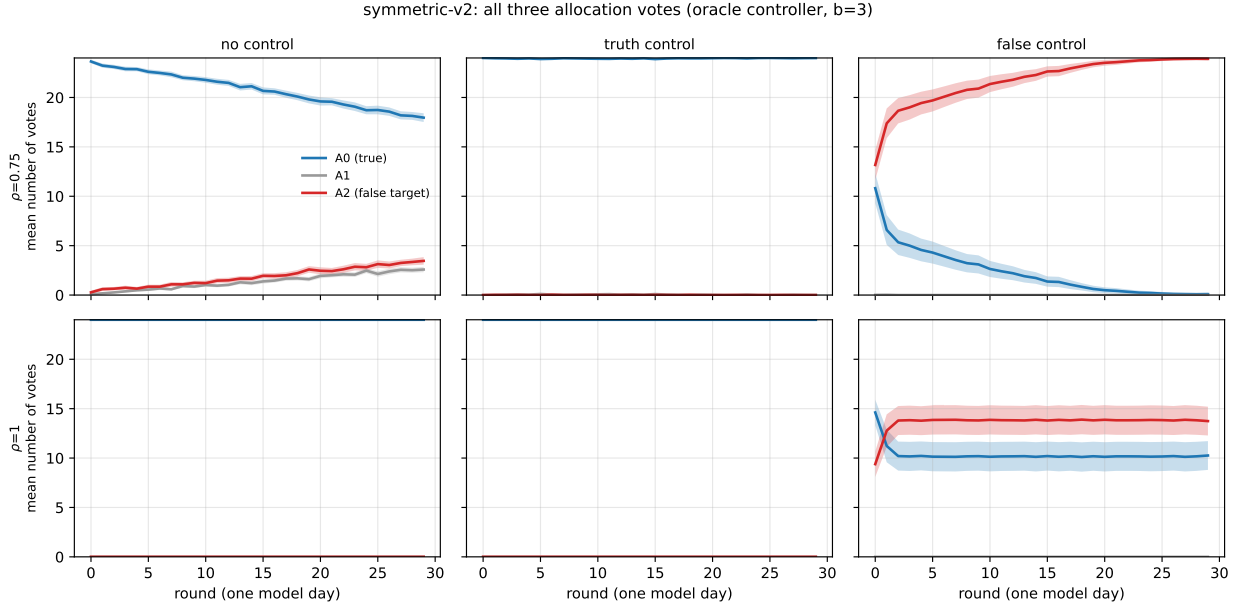


Figure 3: symmetric-v2: the same. In the no-control panels the population finds the right answer on its own; the false controller takes it away even though the agents jointly *own* a proof of `ALLOCATION_0`.

enters at all. Measured: the gap between the truth arm and the false arm is **exactly zero** in all four cells, and the mutual information between the configured target and the coarse state is **0.000 bits**, against **1.000** for a controller that actually looks at its target.

Reading the whole board destroys the randomness. At $\rho = 1$ with `sampling: full` every one of 60 episodes ends *identically*. This reverses our own earlier advice; `uniform` with $q = 7$ is correct.

6 Relation to the task_004 thermodynamics report

A correction. We previously wrote that the task_004 paper’s agent posterior $(28/38, 0, 10/38) = (0.737, 0, 0.263)$ “is exactly our nosolution”. **That is wrong.** It is `task_004` itself — Darius’s original 15-agent setup. **Our** nosolution-v2 is a redesign with a different fact set and 24 agents, and its pooled belief is $(0.500, 0, 0.500)$: a coin flip, by construction. The two are different objects and we had conflated the name “nosolution” with our specific redesign.

What *is* exactly shared: the paper’s controller pool $\mathcal{F}_2$, nine facts with pooled belief $(0.0668, 0, 0.9332)$, is precisely the `ALLOCATION_2`-leaning subset of Darius’s balanced 27-fact pool. Verified.

	task_004 paper	our simulation
purpose	stochastic thermodynamics of feedback	mirror of the MA-CC runtime
time	continuous, Poisson thinning	discrete rounds
memory	three states: absent / passive / active	two: known / active
erasure	yes	none
vote	softmax on probabilities, $\beta = 20$	arg max, no parameter
controller pool	9 facts, target-specific	12 facts, neutral and shared
horizon / agents	600–1000 days / 15	30 rounds / 24
free parameters	nine rates	none

The reconciliation. The paper’s Appendix C identifies the *same* least-recently-used ranking we did. It is not harmed by it, because *its pool is target-specific*: all nine facts favour `ALLOCATION_2`,

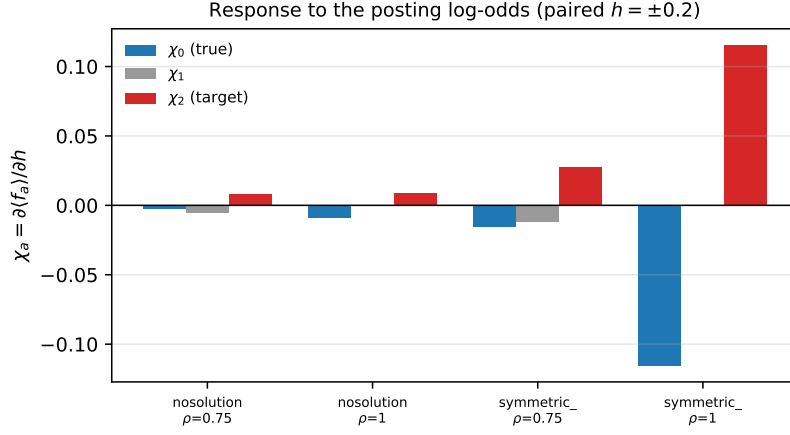


Figure 4: Response of the three vote shares to a small change h in the controller’s posting log-odds, measured as a paired finite difference at $h = \pm 0.2$, as in the task_004 report. The three responses sum to zero by construction. They are the same order of magnitude as the published values ($\chi_2 \simeq 0.019$ there).

so the steering lives in the pool rather than in the ranking. Our design moves the steering into the ranking — and the ranking cannot carry it. Both findings are correct for their own design.